

A Survey on Security Attacks in Electronic Healthcare Systems

R. Priya, S. Sivasankaran, P. Ravisasthiri and S. Sivachandiran

Abstract—Electronic health care record is a growing patient record model that are maintained and stored by third party by which the health record of a patient can be exchanged between doctors, specialist for providing better treatment. At the time of exchanging information security places a vital role in maintaining the patient's data. As the health care system are connected into the network through internet the system is exposed to many security and cyber-attacks. A good security system must not compromise the fundamental security elements such as authentication, confidentiality, integrity, availability and non-repudiation.

Index Terms—Electronic health care, security attacks, genomic, security issues, data leakage, network security

I. INTRODUCTION

A new enactment of digitalized patient's record, managing electronic health record must leads to data security there should not be any security breaches to the patient's data. Exchanging of health information becomes a common process though it is shared process it must be properly linked and maintained properly at the same time the patient's privacy should not be trickled. [1,2] Nowadays, maintaining the patient's health information confidentially is a challenging tasks due to the technological growth and some of the vulnerabilities present in the system. The main objective of the healthcare system is to provide health information at anytime, anywhere to the authentic persons. Security plays a vital role in accessing these information. [3-4] The basic security requirements that every system must include authentication, confidentiality, integrity, availability, non- repudiation. Privacy is considered to be a most important sensitive aspect

on the regard of patient's health record. There should not be any disclosure of health record to the unauthorized persons. [5-7] Also, there must be a confident communication among the patient and specialist. This paper aims to provide the various security attacks that are exists in health care systems. The organization of the paper are as follows Section I contains the Introduction, Section II covers the Literature review, Section III comprises the security requirements, Section IV mention about the Security attacks, Section V gives the conclusion of this paper respectively.

1.1 GENOMIC CHARACTERISTICS

A. Uniqueness

Anticipating for alike twins, each of them has their own distinctive genetic/ genomic pattern. Accordingly, autonomously gathered examples can be coordinated with generally high certainty on the premise of a little number of genomic variants. Consolidated databases of hereditary/genomic data could conceivably be excavated for individual distinguishing proof purposes. Moreover, as logical comprehension of the relationship between gene sort and phenotype increments, hereditary/genomic data might be utilized all the more precisely to foresee an individual's physical qualities from his/her DNA series data.

B. Predictive capability

Some hereditary/genomic tests can anticipate the probability of building up a given infection or the reaction to a particular medication. The unpredictable interrelationships inside an individual's genome also, how it responds to natural conditions at last denies what may really happen. The prescient way of hereditary qualities is a basic yet complex thought for creating arrangement as to/genomic test data. For instance, while this data can advise pre-emptive activity, it might additionally be utilized to separate in view of inclination. In addition, hereditary/genomic test results could be utilized as a part of the nonattendance of other supporting clinical signs or side effects to advise social insurance administration choices.

C. Immutability

Hereditary data is additionally unvarying; a person's germline/acquired data does not change all through life. (A few cells in the body may have a change in DNA after origination, for example, those presented amid DNA replication, for instance. These changes, alluded to as physical mutations, may cause tumour or different sicknesses. Acquired

R. Priya is with the Information Technology Department, RAAK College of Engineering and Technology, Puducherry. (e-mail: priyatechit20@gmail.com).

S. Sivasankaran is with the Information Technology Department, RAAK College of Engineering and Technology, Puducherry. (e-mail: sankarsvision@gmail.com).

P. Ravisasthiri is with the Information Technology Department, RAAK College of Engineering and Technology, Puducherry, India. (e-mail: ravisasthiri.p@gmail.com).

S. Sivachandiran is with the Computer Science and Engineering Department, RAAK College of Engineering and Technology, Puducherry. (e-mail: sivachandiran.s@gmail.com).

DNA does not generally change, and these physical transformations cannot be acquired or passed on). Thus, open exposure of individual hereditary/genomic test data could make long enduring and unusual impacts, given unexpected innovative and interpretive advances.

D. Requirement of testing

Numerous hereditary markers, especially those for ailment inclination and medication reaction, can't be discovered in the ordinary course of clinical care; they should be gotten from a hereditary/ genomic test. In spite of the fact that appearances of a few illnesses (e.g., Huntington infection) infer the nearness of certain hereditary changes, testing is normally required to advise a person of a particular change that inclines him or her to a particular condition.

E. Historical misuse

Hereditary data has been abused to advance genetic counselling initiatives, segregate in protection and the work environment, and acquire data about people's therapeutic histories. As hereditary research and therapeutic applications propel, the capacity to relate hereditary inclination for sickness to elements, for example, gender, self-recognized race, or inhabitants gathering will probably in wrinkle. In spite of the fact that populace based research advises epidemiologic request, concerns flourish that hereditary/genomic test in-development could be utilized improperly to generalization or demonize people.

F. Variability in public knowledge and perspectives

There is wide inconstancy in individual comprehension about the part of hereditary qualities in fitness and aliment, individual susceptibility concerning/genomic test data, and sentiments about hereditary qualities (e.g., ontological contemplation in light of hereditary reductionist).

G. Impact on family

Hereditary/genomic test data likewise can possibly affect an individual's relatives, as germline changes i.e., changes contained in the sperm or egg that might be gone to posterity) may uncover data about restorative dangers to blood-relatives. In this way, an individual's choice to experience a hereditary/genomic test could uncover data that recommends hazard to relations with respect to the potential advancement of an unending or incapacitating issue.

H. Temporality

Societal viewpoints and the capacity to translate hereditary/genomic test data will probably develop after some time, as will strategies with respect to the utilization of such data in social insurance basic leadership. For instance, a contemporary test that groupings a particular quality may yield understanding into the danger of building up a specific infection. In any case, in time this grouping in- arrangement may demonstrate useful with respect to various conditions and additionally restorative responses. Given the exponential development of the field of sub-atomic diagnostics, the extension and capacity to translate test results will advance quickly. In parallel, expanding open mindfulness about the potential utilization of hereditary/

genomic data to social insurance choices will impact points of view with respect to how to manage this data.

I. Ubiquity and ease of procurement

Hereditary material is anything but difficult to get. DNA can be acquired from saliva, blood, hair, and different tissues that are kept on a surface. Along these lines, an individual's genomic data can be promptly got without his/her insight or consent. The attributes talked about here must be considered comprehensively when advising arrangement on the utilization of hereditary/genomic test data in patient fitness records. Despite the fact that a solitary trademark may not bolster a contention for or against treating hereditary/genomic test data as "outstanding," the consolidated impact of these attributes may impact matters of potential exception. A coordinated system ought to in this way be utilized to survey the full effect of arrangement choices.

II. RELATED WORK

Alevtina Dubovitskaya [8], they addressed the difficult of building an anonymized medical database from multiple sources. For these problem author suggested the solution, to achieve data integration in a heterogeneous network of many clinical institutions, while preserving data utility and patients' privacy. Firstly, the author proposed a secure and scalable cloud eHealth architecture to store and exchange patient's data for the treatment. Secondly, present RSBD algorithm for efficient aggregation of the health data for the research purposes from multiple sources independently.

Salah Bindahman and Nasriah Zakaria [9], they discuss the overall privacy concept in Health Information System (HIS), since most HISs are online based. The unlikeness between privacy, security and some related works in HIS and suggestions on user can protect their privacy are discussed.

Farzana Rahman [10], proposed 'PriGen', it is a generic framework that preserves the privacy of sensitive healthcare data in the cloud. PriGen allows the users to preserve privacy while accessing cloud based healthcare service without the assistance of a third party. With the use of homomorphic encryption function on sensitive private information; this framework maintains confidentiality of private information sent by the cloud users to untrusted cloud based healthcare service providers.

Mohd Anwar [11], aims to systematically rationalize and explore security-privacy related issues in providing anywhere, anytime healthcare services. They survey the existing approaches and discuss health IT infrastructural governance, institutional and cross-national policy challenges to address the relevant security and privacy issues and categorize these issues in relation to the users, applications, communications, and devices. A consolidated effort from technological, human factor, and social research communities can lead to an adequate response to key privacy-security issues in this nascent anytime, anywhere healthcare paradigm.

Lina A.Abuwardih [12], illustrated a survey for the privacy models for discloser attack, membership discloser attack and attribute discloser attack. And also discussed the list of models and techniques in various discloser attacks that are used for publishing data about patients.

III. SECURITY REQUIREMENTS IN HEALTHCARE

Any system must accomplish the security accompanying desires in order to safeguard the information in a reliable manner.

A. *Authentication*: Authentication is a procedure by which a framework checks the character of a user who wishes to get to it. It is considered to be a powerful security.

B. *Integrity*: It confirms during transmission no data or information are not altered or modified by the attacker. The system must provide 100% data integrity thereby the trust level will be increased.

C. *Confidentiality*: It confirms that no information are illegally accessed by the attacker.

D. *Availability*: It must provide access to the information whenever it is required by the legitimate users. There should not be any delay in information to the authentic persons.

E. *Non-Repudiation*: Once the information are sent by the sender will not be able to deny the information that is sent.

IV. SECURITY ATTACKS IN HEALTHCARE SYSTEM

Healthcare systems are susceptible to security attacks which are executed intentionally or unintentionally to grab the profit or to control the entire system. This degrades the performance of the healthcare systems.

The network, patient's health records that are maintained in the hospital can be attacked or embezzled by malicious hackers by this the trust level is dropped between the healthcare care system and the users.

A. Security attacks at data gathering phase

1) *Key logger Attack*: The malicious attacker runs the key logger into the system to capture the keystrokes. Based on the keystrokes attacker obtains the credentials for illegally access of patient's health information.

2) *Phishing Attack*: The attacker sends an illegitimate email link to the patient to update their health information. If the patient who are all unaware of this attack may fall in attacker's trap. The information obtained can be given to the opponent for money laundering or blackmailing.

3) *Social Engineering Attack*: In this, the third party may impersonate as legitimate by tricking the system to grab the patient record.

4) *Log forgery*: The patient, doctors, nurse have their own credentials to access the health care system. Normally, they follow role based access unfortunately if anyone of the

credentials which belongs to the higher officials are known they can make a forged access to the system and can make data leakage, data alterations, or exploring patient's health information illegally.

5) *Brute force Attack*: It involves trial and error method for guessing the password. This attack becomes very simple when the attacker knows entire information about the opponent.

B. Security attacks at network phase

1) *Man in the middle*: It is of the common type of attack that occurs in the network in which the attacker may impersonate into the network and may gain access to the information that are taking place between patient and the database or server. The attacker sniffs the entire communication that occurs in the network.

2) *Eavesdropping of patient's health report*: It is an unlawful interception of communication in the network. If anyone is accessing the patient's record legally the attacker sniffs the communication illegally.

3) *Pharming*: This attack is related to phishing attack. The malicious codes are installed in the victim's system by which the malicious website are executed by compromising the DNS server. It mainly redirects to the malicious website.

4) *Denial of service*: It is a common type of attack in the network in which the attacker captures the entire network by flooding unwanted traffic delays to the health care system. The legitimate users cannot be able to access the health care information due to this flooding traffic.

5) *Data Tampering*: The records are altered by the attacker by mimicking the attacker as a legitimate. It includes illegal modification or may cause damage to the health record.

6) *Scrambling Attack*: It is a type of jamming attack in which the attacker can trigger the WiMax frames at small period of times. Attackers may distinguish the network to scramble the entire typical operation of the network.

7) *Dumpster diving*: It is a type of search through dumpster to find the information related to company employee, infrastructure of a network, product information.

8) *Wiretapping Attack*: The attacker tampers the network through telecommunication devices to eavesdrop the phone call communication.

9) *Message modification*: After capturing the network the attacker alters the patient's record by which it creates mislead in the treatment of ailments.

10) *Data interception*: This attack occurs when transferring patient's health information to the computer system that are connected via the hospital network.

11) *Spoofing Attack*: The malicious attacker mimics as a legitimate person to enter into the network and controls the system.

C. Security attacks at storage phase

1) *Password Attack*: They are executed to crack the particular patient's health record or to gain access and to control the entire healthcare system.

2) *Cross site scripting Attack*: In this the attacker injects some malicious scripts to gain the access for healthcare system.

3) *Data diddling Attack*: Illegal access or unauthorized access to the system which leads to data alteration thereby it modifies the patient's health record.

4) *Weak Authentication Attack*: The data that are present are accessed illegally by exploring vulnerability that are exist in the system. It act as a backdoor to the attacker to gain the access.

5) *Targeted data mining Attack*: It applies some data mining techniques to obtain the access for database to seize the sensitive information of the patient.

6) *SQL Injection Attack*: It is a type of injecting the malicious SQL statements to the database inorder to explore the vulnerability that are present by which the health record can be accessed by the attacker.

7) *Aggregation Attack*: It focusses on non-sensitive piece of information. Finally all the non-sensitive information are combined to acquire the sensitive information from the patient's health record.

8) *Inference Attack*: It is described by the attack to penetrating into the database inorder to find out the sensitive information of patient's health record. It is a challenge to defend against the inference attack because the insignificant piece of information can be obtained by authentic access to the database record.

V. CONCLUSION

Electronic healthcare is attaining its attractiveness due to the remarkable growth of technology and smartness towards the internet. Even though the works are smartly shared and maintained security plays a major role in protecting the individual's credentials.

This paper mentions the security requirements for developing the secured health care systems. Also, this paper covers the possible security attacks that are executed in online.

REFERENCES

- [1] José Luis Fernández-Alemán, Inmaculada Carrión Señor, Pedro Ángel Oliver Lozoya, Ambrosio Toval, Security and privacy in electronic health records: A systematic literature review, *Journal of Biomedical Informatics*, Volume 46, Issue 3, June 2013, Pages 541-562.
- [2] Mete Akgün, A. Osman Bayrak, Bugra Ozer, M. Şamil Sağiroğlu, Privacy preserving processing of genomic data: A survey, *Journal of Biomedical Informatics*, Volume 56, August 2015, Pages 103-111.
- [3] <http://www.darkreading.com/attacks-breaches/major-cyberattacks-on-healthcare-grew-63--in-2016/d-d-id/1327779>
- [4] <http://securityaffairs.co/wordpress/54666/security/healthcare-cyber-security.html>
- [5] N. M. Shrestha, A. Alsadoon, P. W. C. Prasad, L. Hourany and A. Elchouemi, "Enhanced e-health framework for security and privacy in healthcare system," 2016 Sixth International Conference on Digital Information Processing and Communications (ICDIPC), Beirut, 2016, pp. 75-79.
- [6] X. Wang and H. Tang, "Privacy Protection in Sharing Personal Genome Sequencing Data," 2012 *IEEE Second International Conference on Healthcare Informatics, Imaging and Systems Biology*, San Diego, CA, 2012, pp. 143-143.
- [7] R. Ghasemi; M. M. A. Aziz; N. Mohammed; M. H. Dehkordi; X. Jiang, "Private and Efficient Query Processing on Outsourced Genomic Databases," in *IEEE Journal of Biomedical and Health Informatics*, vol. PP, no. 99, pp. 1-1
- [8] Dubovitskaya, Alevtina, Urovi, Visara, Vasirani, Matteo, Aberer, Karl, ICT Systems Security and Privacy Protection: 30th IFIP TC 11 International Conference, SEC 2015, Hamburg, Germany, May 26-28, 2015, Proceedings "A Cloud-Based eHealth Architecture for Privacy Preserving Data Integration", Hannes Federrath,, Dieter Gollmann Eds. Springer International Publishing, 2015, pp. 585-598.
- [9] Salah Bindahman, Nasriah Zakaria, Informatics Engineering and Information Science: International Conference, ICIEIS 2011, Kuala Lumpur, Malaysia, November 14-16, 2011, Proceedings, Part IV, "Informatics Engineering and Information Science: International Conference, ICIEIS 2011, Kuala Lumpur, Malaysia, November 14-16, 2011, Proceedings, Part IV, "Privacy in Health Information Systems: A Review", Azizah Abd Manaf, Shamsul Sahibuddin, Rabbiah Ahmad, Salwani Mohd Daud, Eyas El-Qawasmeh Eds. Springer Berlin Heidelberg, 2011, pp 285-295.
- [10] Farzana Rahman, Sheikh Iqbal Ahamed, Ji-Jiang Yang, Qing Wang, nclusive Society: Health and Wellbeing in the Community, and Care at Home: 11th International Conference on Smart Homes and Health Telematics, ICOST 2013, Singapore, June 19-21, 2013. Proceedings, "PriGen: A Generic Framework to Preserve Privacy of Healthcare Data in the Cloud", Jit Biswas, Hisato Kobayashi, Lawrence Wong, Bessam Abdulrazak, Mounir Mokhtari Eds. Springer Berlin Heidelberg, 2013, pp. 77-85.
- [11] Mohd Anwar, James Joshi, Joseph Tan, Anytime, anywhere access to secure, privacy-aware healthcare services: Issues, approaches and challenges, *Health Policy and Technology*, Volume 4, Issue 4, 2015, Pages 299-311.
- [12] Lina A. Abuwardih, Wa'ed Shatnawi, Ahmed Aleroud, "Privacy preserving data mining on published data in healthcare: A survey", 2016, pp. 1-6.